

Optimally managing threats to biodiversity across large scales

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Environment and
Climate Change Canada



Conservation is actions in places

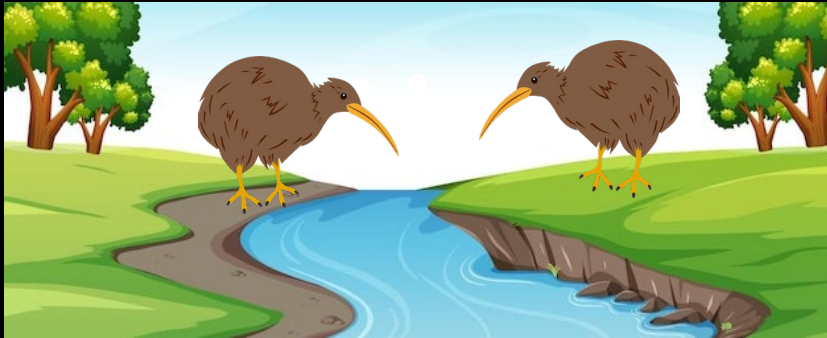
1. Area impacted by threat



2. Implement action to manage threat



3. Positive conservation outcome



Which threats to manage, and where?

Current

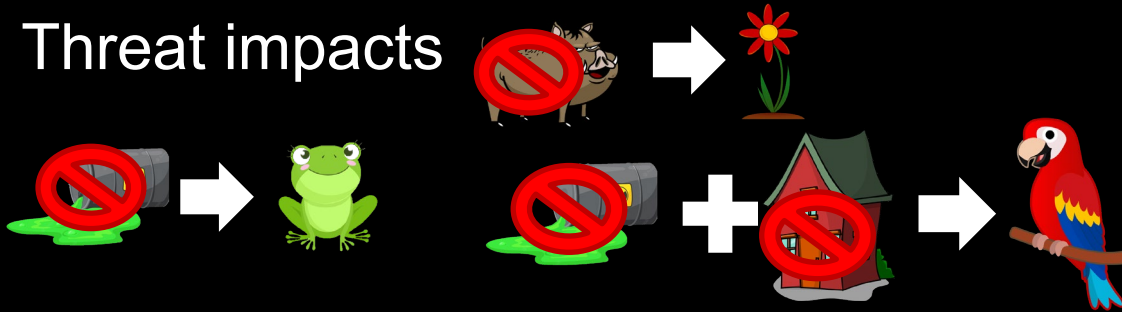


Which threats to manage, and where?

Current



Threat impacts



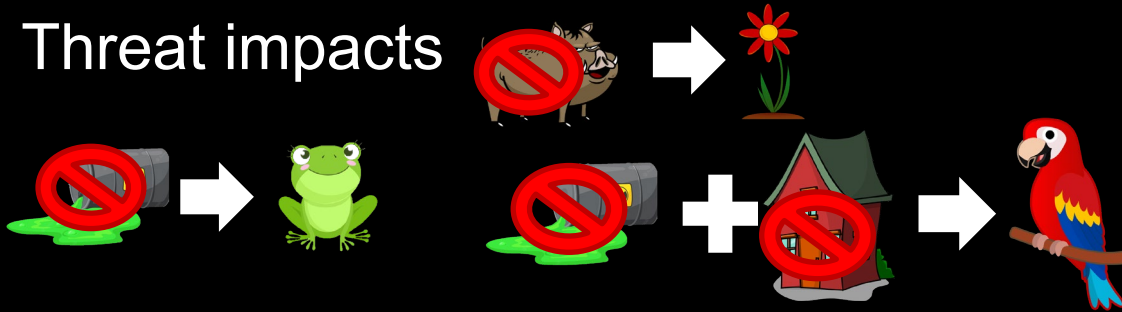
Which threats to manage, and where?

Current

Potential



Threat impacts



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Current



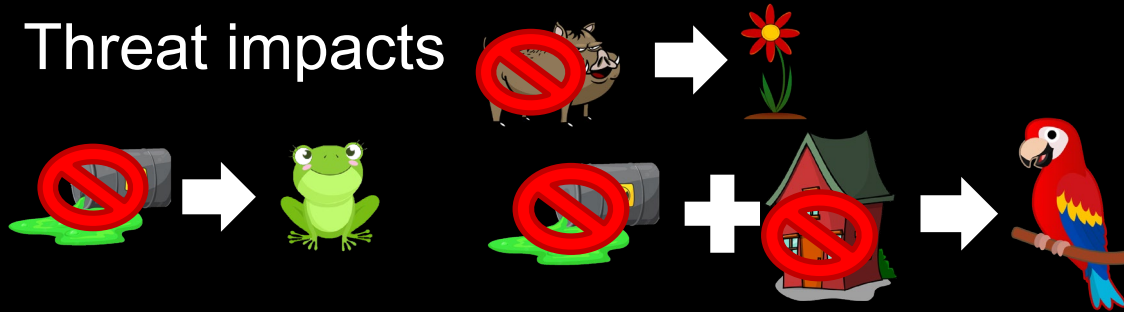
Potential



Priority



Threat impacts



Targets



Ensure each species has enough of “threat free” habitat

Current



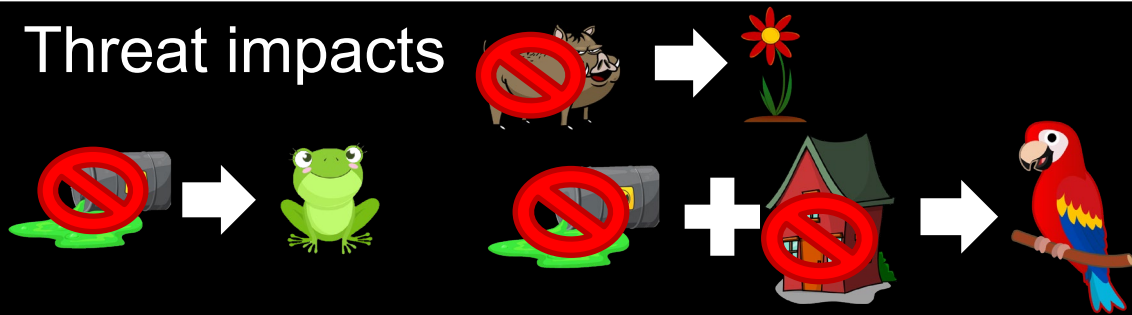
Potential



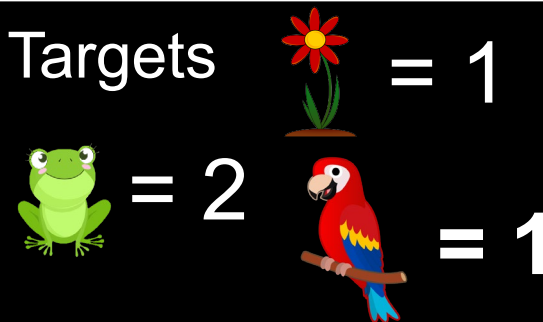
Priority



Threat impacts



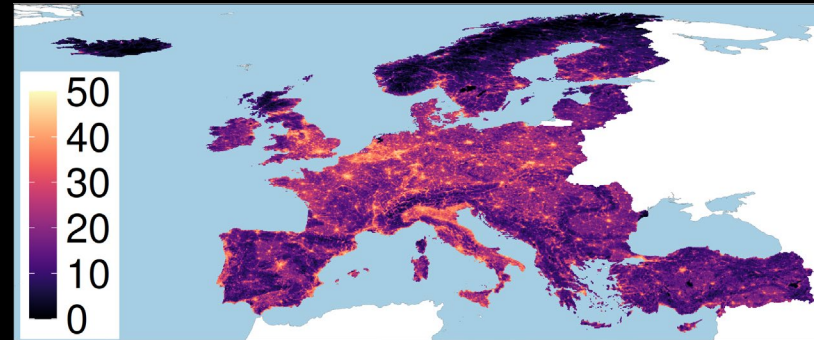
Targets



European case study

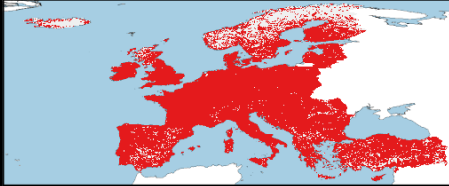
- 710 species: 81 amphibian, 378 birds, 156 mammal, 95 reptile species
- 165,000+ planning units (grid cells)
- Natura 2000 network, nationally designed protected areas, OECMs
- Conservation benefit for a species = amount of threat-free habitat in conservation areas

Human pressure

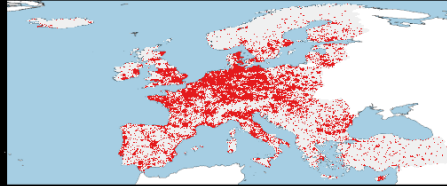


Threats to biodiversity

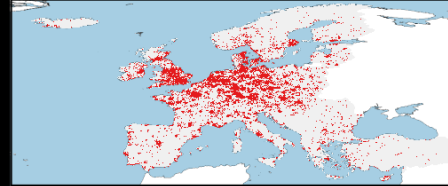
1.1: Housing & urban areas



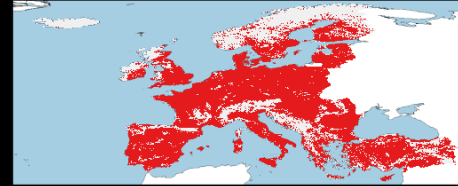
1.2: Commercial & industrial areas



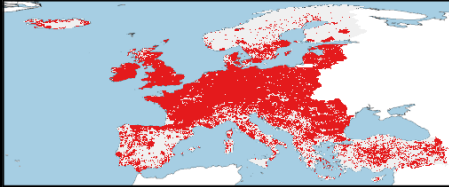
1.3: Tourism & recreation areas



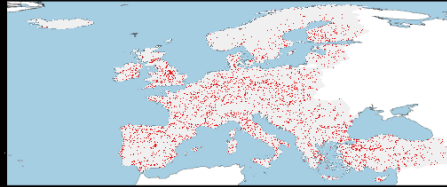
2.1: Annual & perennial non-timber crops



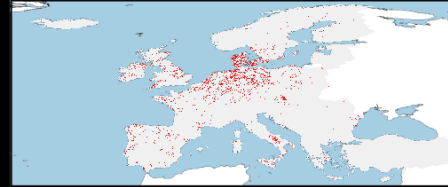
2.3: Livestock farming & ranching



3.2: Mining & quarrying



3.3: Renewable energy



4.1: Roads & railroads



5.1: Hunting & collecting terrestrial animals



5.2: Gathering terrestrial plants



9.4: Garbage & solid waste



Threats to biodiversity

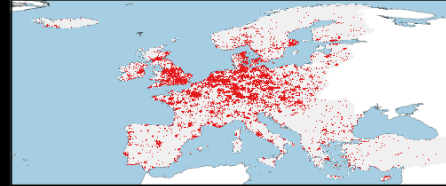
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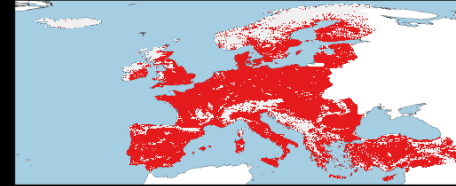
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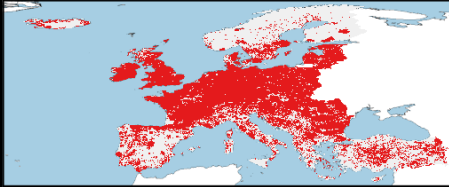
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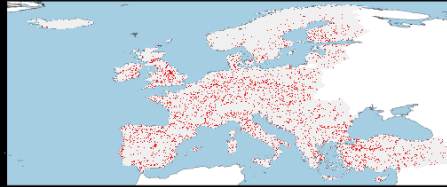
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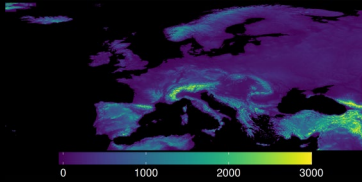


 present
 absent

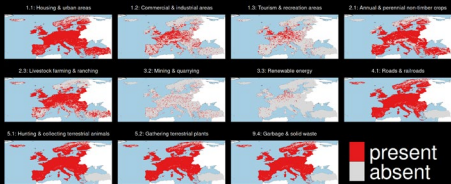
Mapping consequences

Species

Elevation

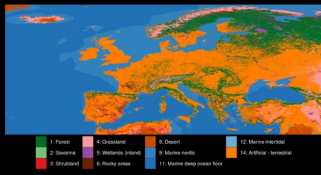


Threats



Current
land cover

Potential natural
vegetation



Horned Grebe



What if?

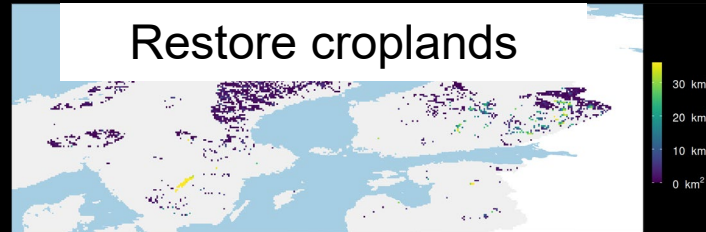
Protect existing habitat



Restore urban areas



Restore croplands

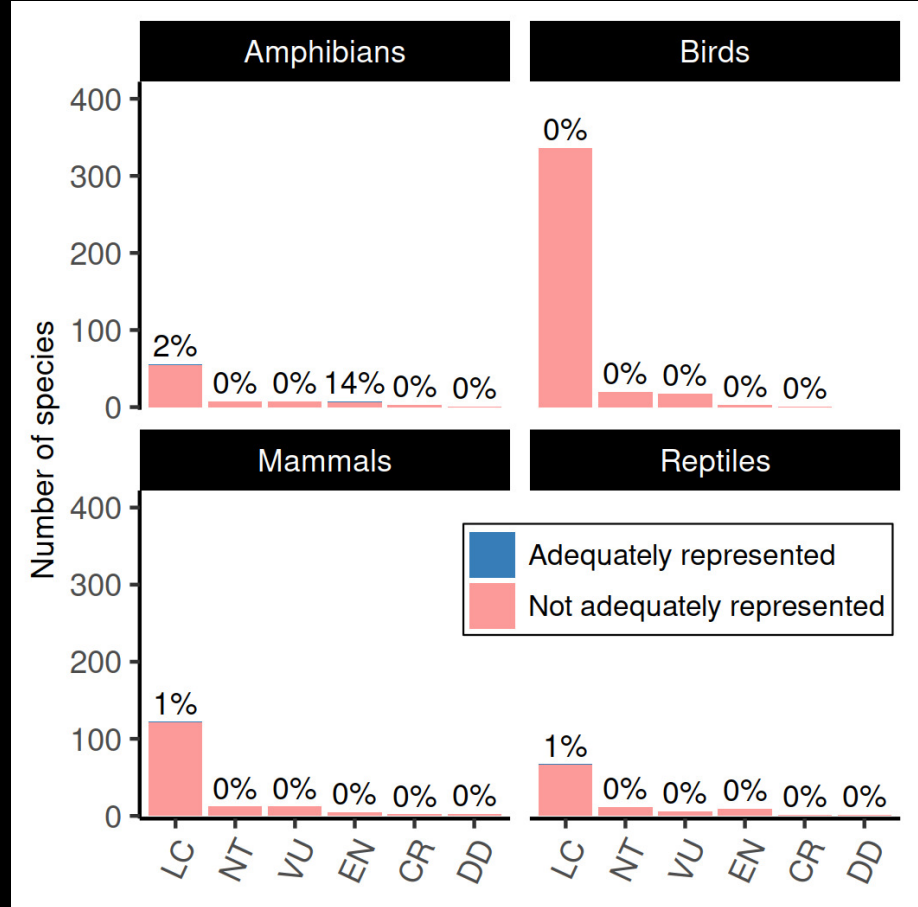


Restore urban & croplands

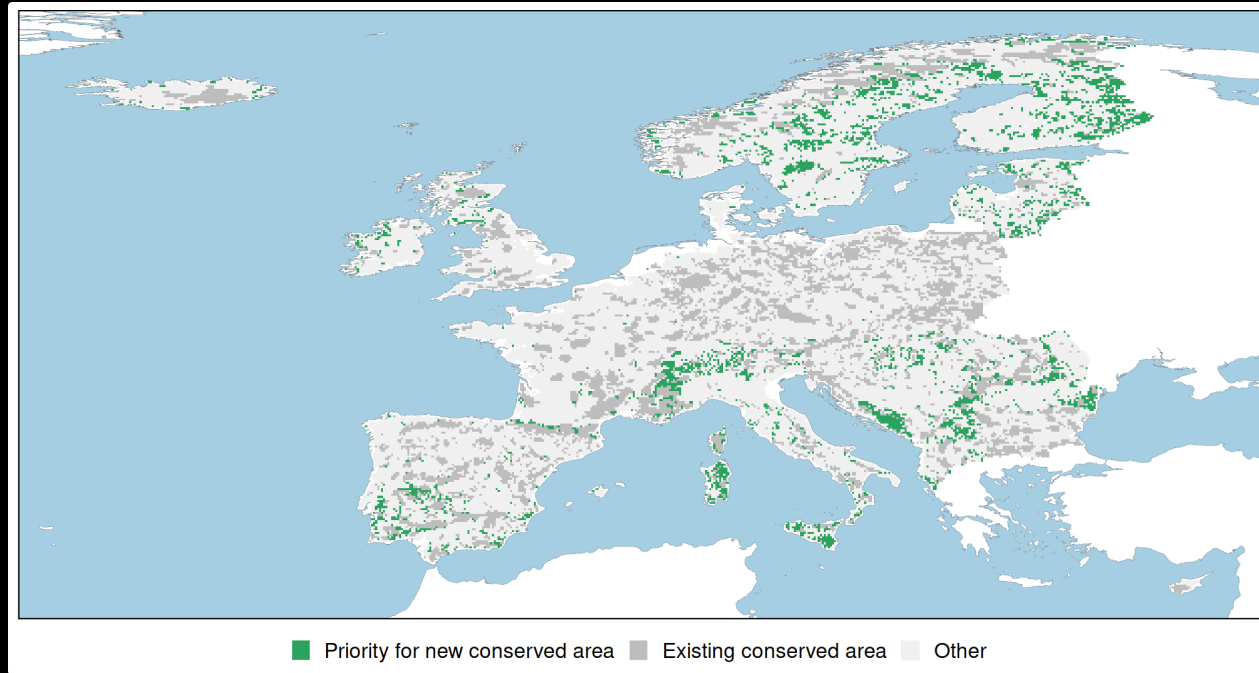


Performance of existing protected areas

- Only 4 / 710 species with adequate threat-free habitat within protected areas
- Only 1 / 74 imperilled species
- Much worse than previous assessments which don't account for threats



Establishing protected areas to cover 30% of Europe



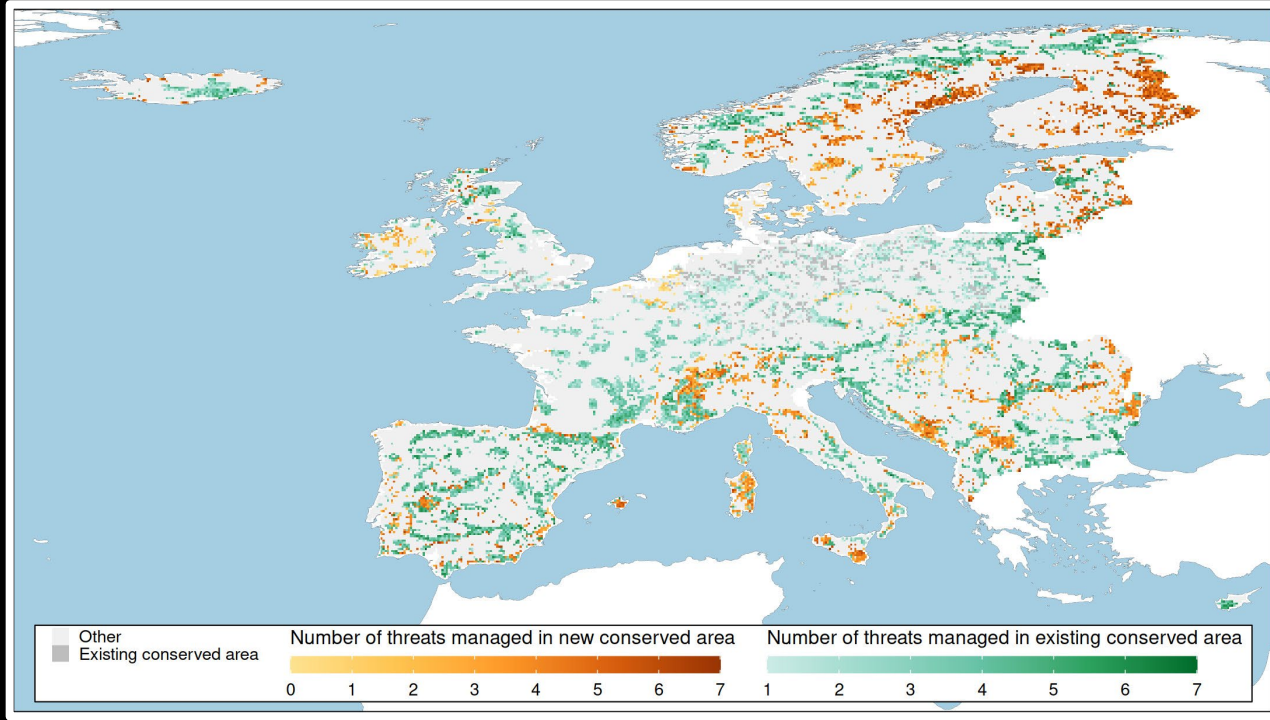
Improves from 4 / 710 to 27 / 710 species with adequate threat-free habitat within protected areas

Managing threats within existing protected areas



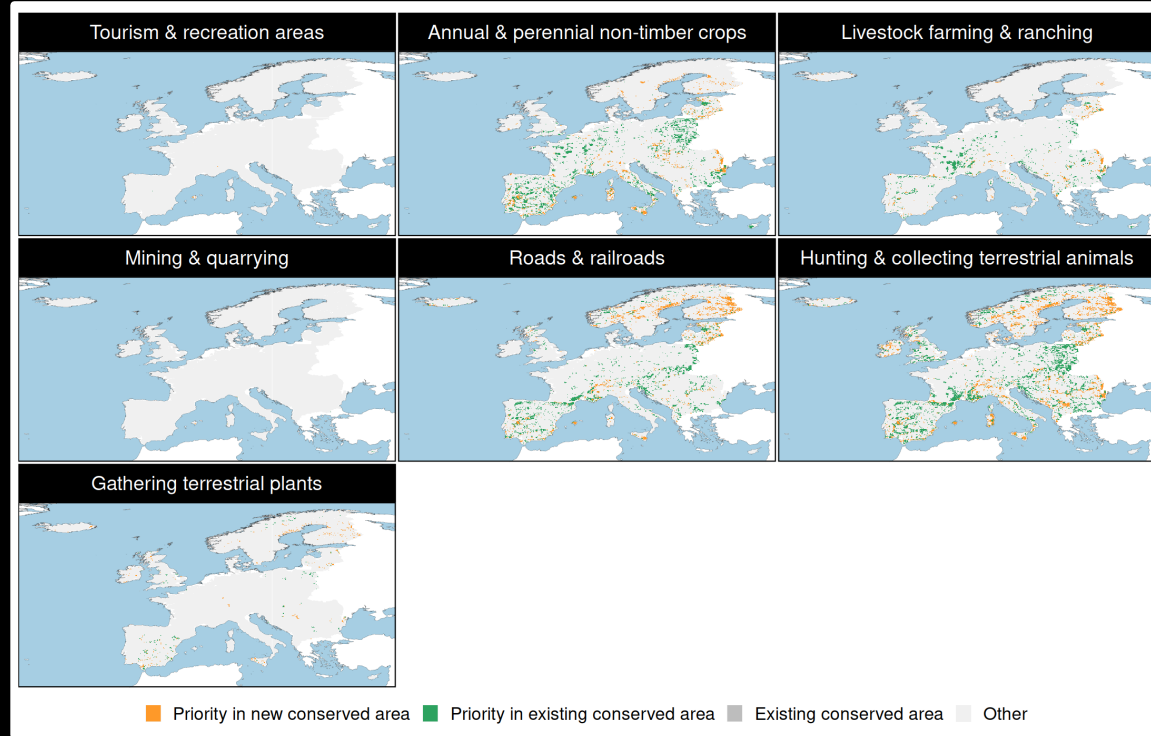
Bigger improvement from 4 / 710 to 113 / 710 species with adequate threat-free habitat within protected areas

Prioritizing threat management across 30% of Europe



Massive improvement from 4 / 710 to 178 / 710 species with adequate threat-free habitat within protected areas

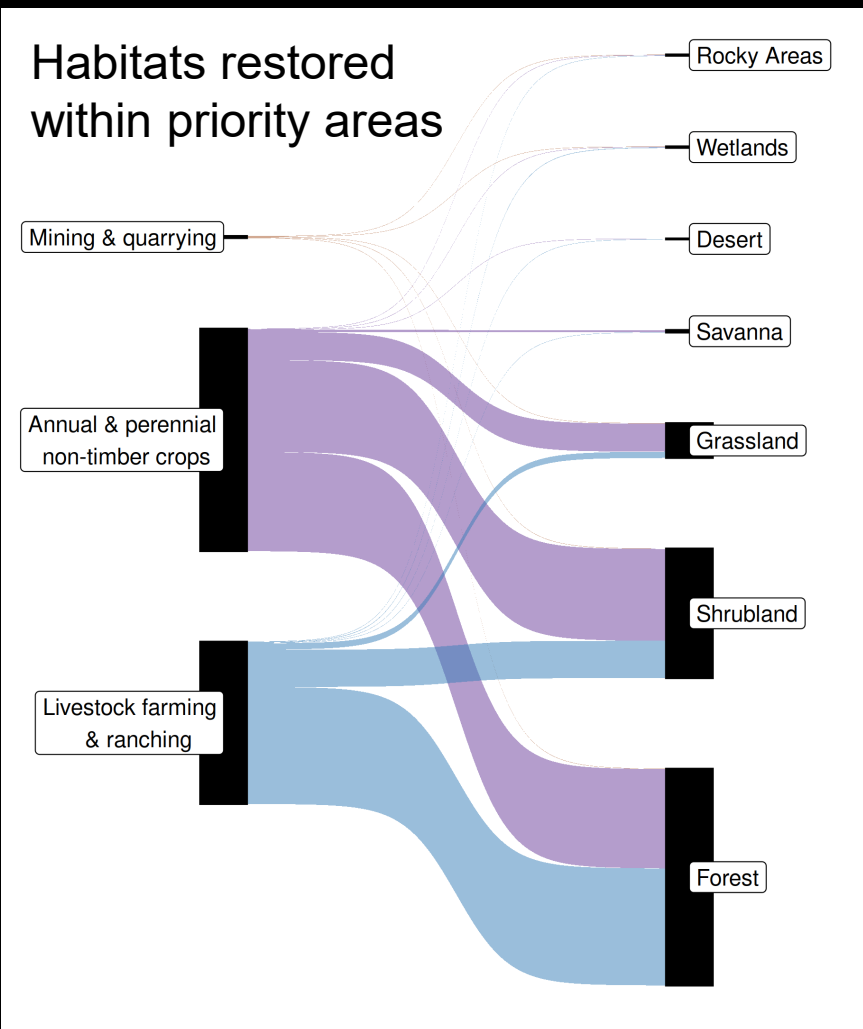
Prioritizing threat management across 30% of Europe



Most management focussed on livestock farming and ranching, hunting and collecting terrestrial animals, and annual and perennial non-timber crops

Restoration through threat management

- Most of the priority areas that involved habitat restoration focussed on providing forest and shrubland areas
- 69% of priority areas involved restoring habitat within existing protected areas
- 30% of priority areas involved restoring habitat within new protected areas



Take home messages

Most EU species don't have adequate habitat in protected areas that is free from threats

Improving management of existing protected areas is necessary, but insufficient, for many species

Strategically managing threats both within and beyond existing protected areas is needed

**Conservation is actions in places:
you need to think about both!**



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